Ghazi University

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Class:- BSCS Sec(E)

Subject:- Programming fundamental.

Submitted to:- Sir Asad.

Lab 5:-

- **Task 1:-**

C++ program to print table of any Number.

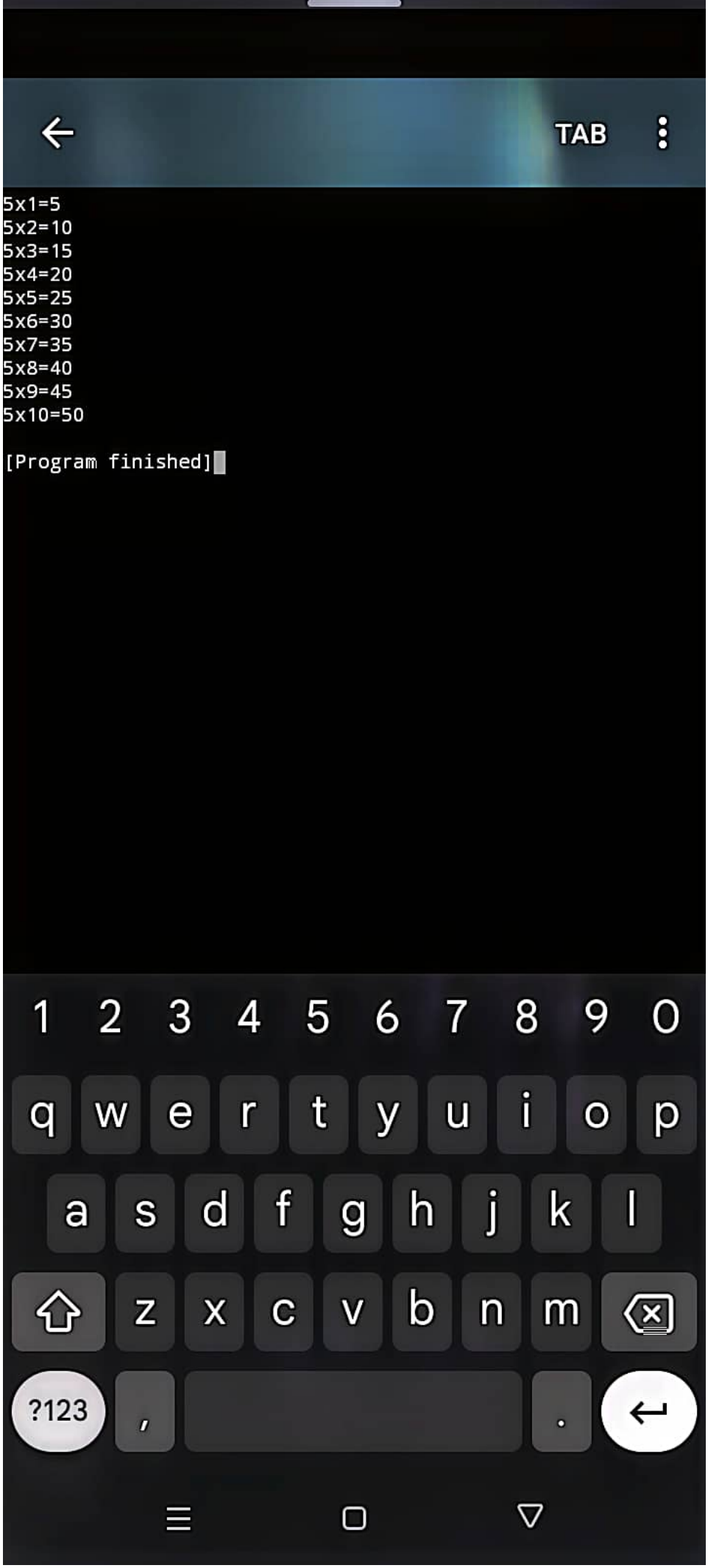
Input:-

```
#include <iostream>
using namespace std;
int main( )
{
int num = 5;
for (int i=1; i<=10; i++)
{
cout << num << "x" << i << "=" << num * i << endl;
}
return 0;
}
```

Output:-

5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

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Task 2:- Find power of any Number using for loop.

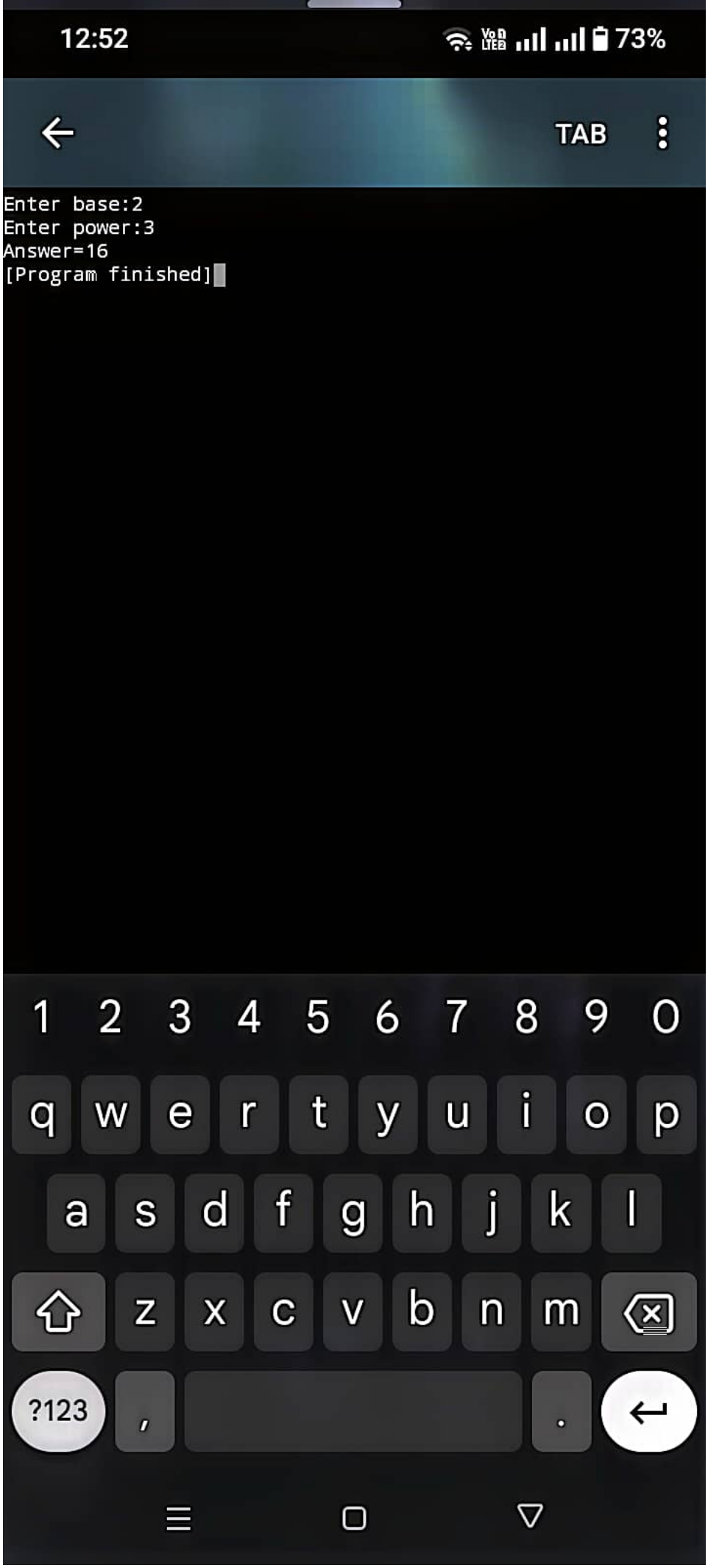
Input

```
# include <iostream>
 using namespace std;
 int main ()
{
int base, power, result = 1;
cout << "Enter base:";
cin >> base;
cout << "Enter power:";
 cin >> power;
for (int i=0; i<= power; i++)
{ result= result * base;
 cout << "Answer = " << result;
}
 return 0;
}
```

Output:

Enter base: 2
Enter power: 3
Answer = 16

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Task 3:-

Sum of Natural Numbers using loop.

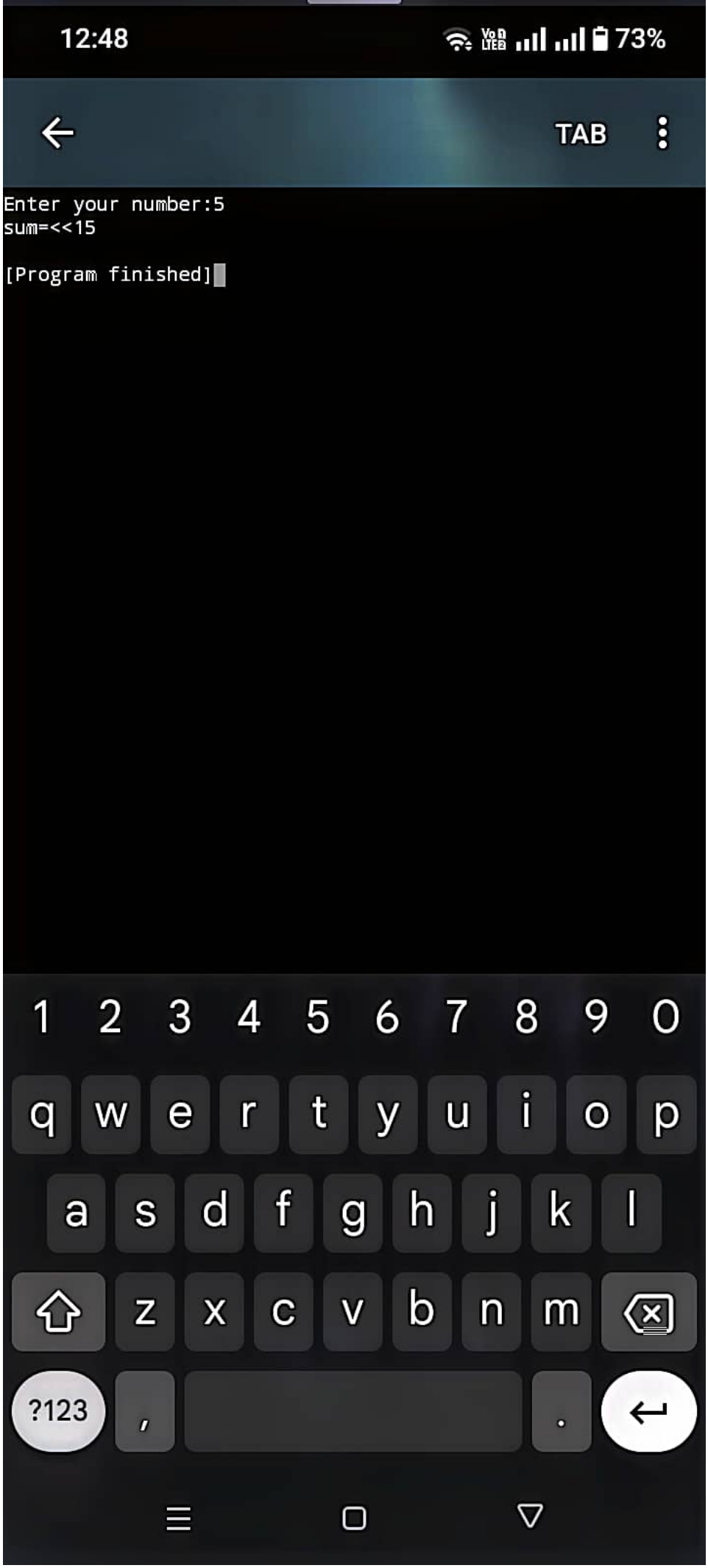
Input :-

```
# include <iostream>
using namespace std;
int main ()
{
    int n, sum = 0;
    cout << "Enter your Number:";
    cin >> n;
    for (int i = 1; i<= n; i++)
    {
        sum = sum + i;
    }
    cout << " sum = <<" " << sum << endl;
    return 0;
    }
```

Output :-

```
Enter your number : 5
sum = << 15
```

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Task 4:-

C++ program to Find Sum of digits of a Number.

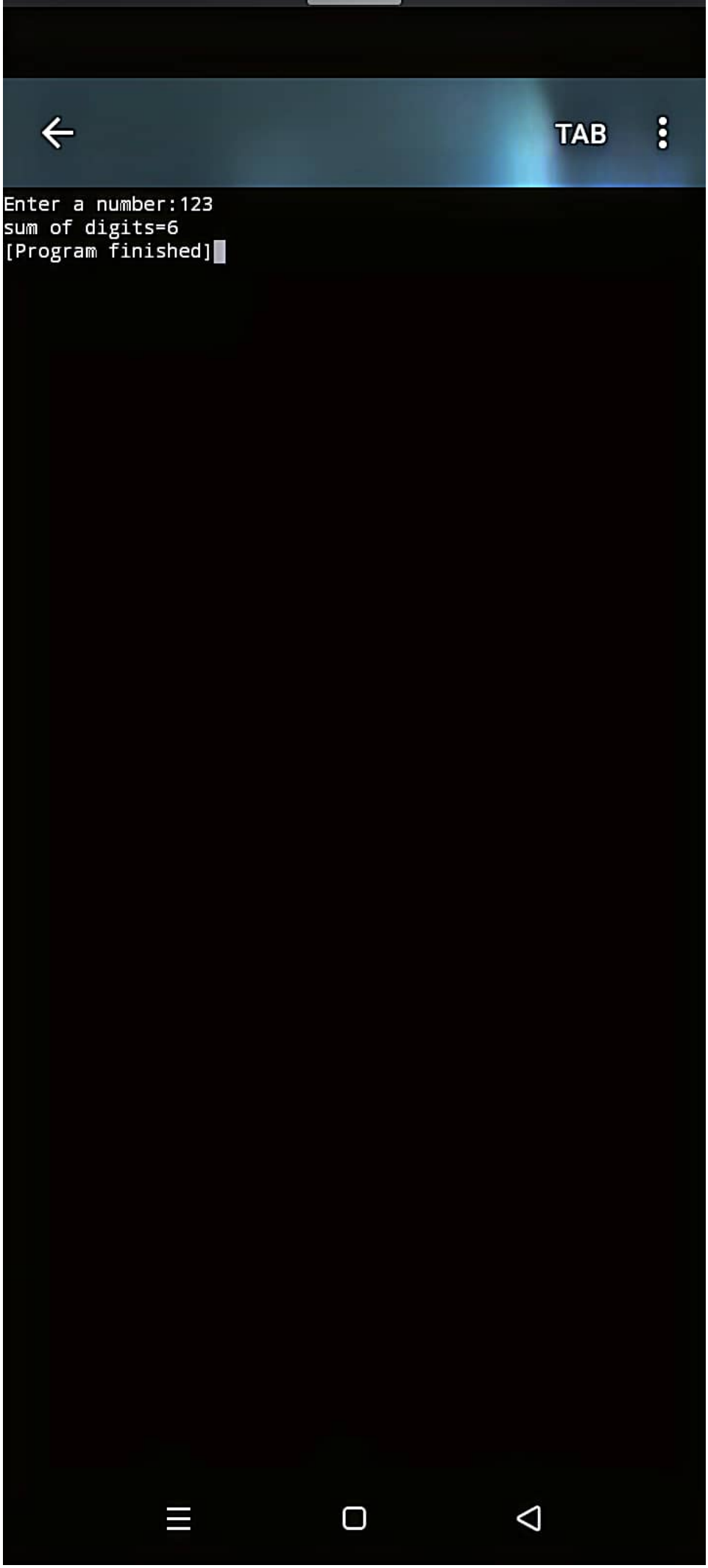
Input:-

```
# include <iostream>
using namespace std;
 int main ()
{
int num, digits, sum=0;
 cout << "Enter a Number:";
 cin >> num;
 while(num>0)
{
 digit = num %10
 sum =  sum +digit;
 num = num /10;
}
cout<<"Sum of digits="<<sum;
 return 0;
}
```

Output:-

Enter a Number = 123
sum of digits = 6.

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Task 5:-

Input:-

Program To Generate factorial a Certain Number.

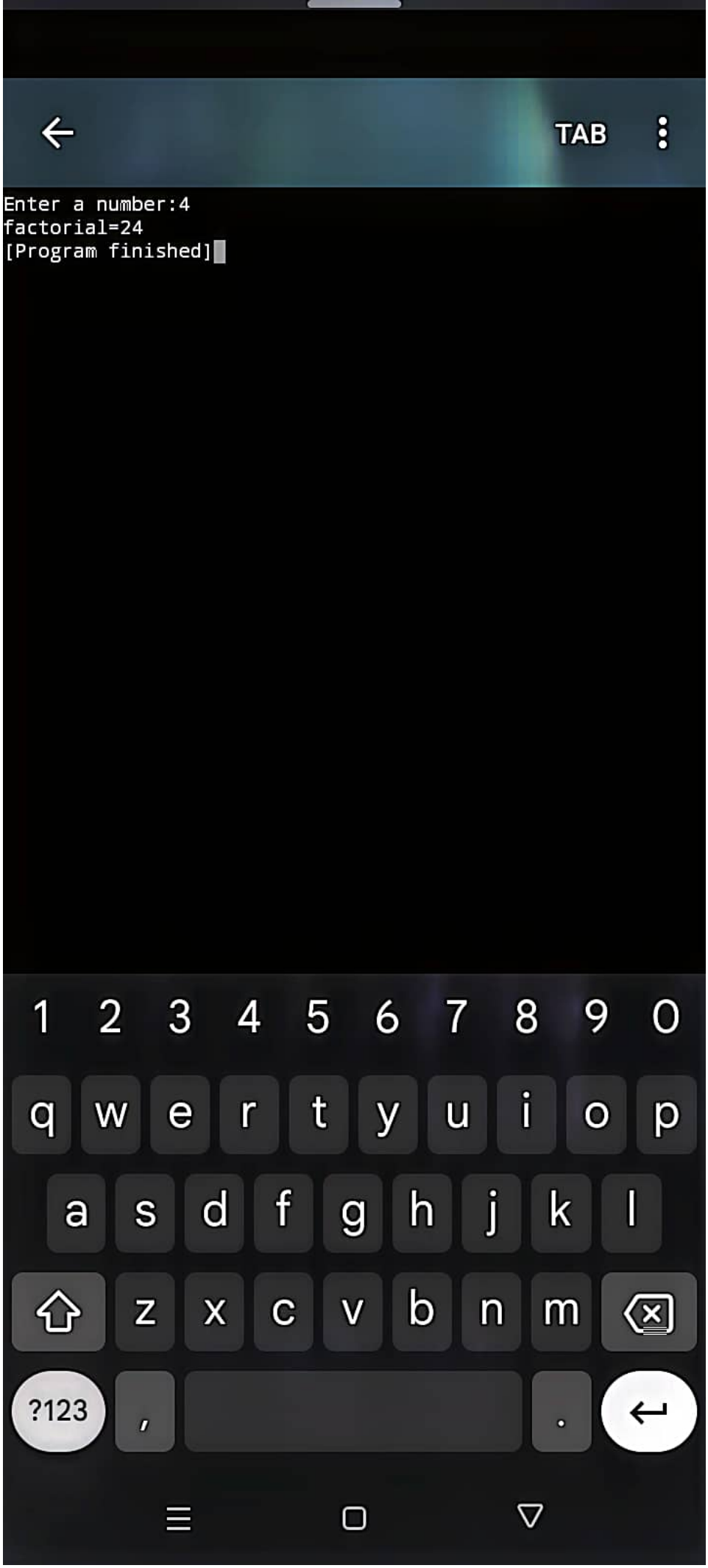
Input:-

```
# include <iostream>
  using namespace std;
  int main ()
{
   int n, fact = 1;
   cout << "Enter your Number:";
   cin >> n;
  for (int i=1; i<=n; i++)
{
   fact = fact * i;
}
  cout << "factorial =" << fact;
  return 0;
}
```

Output:-

Enter a Number: 4!
factorial = 24.

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Task 6:-

C++ program that will print the following pattern.

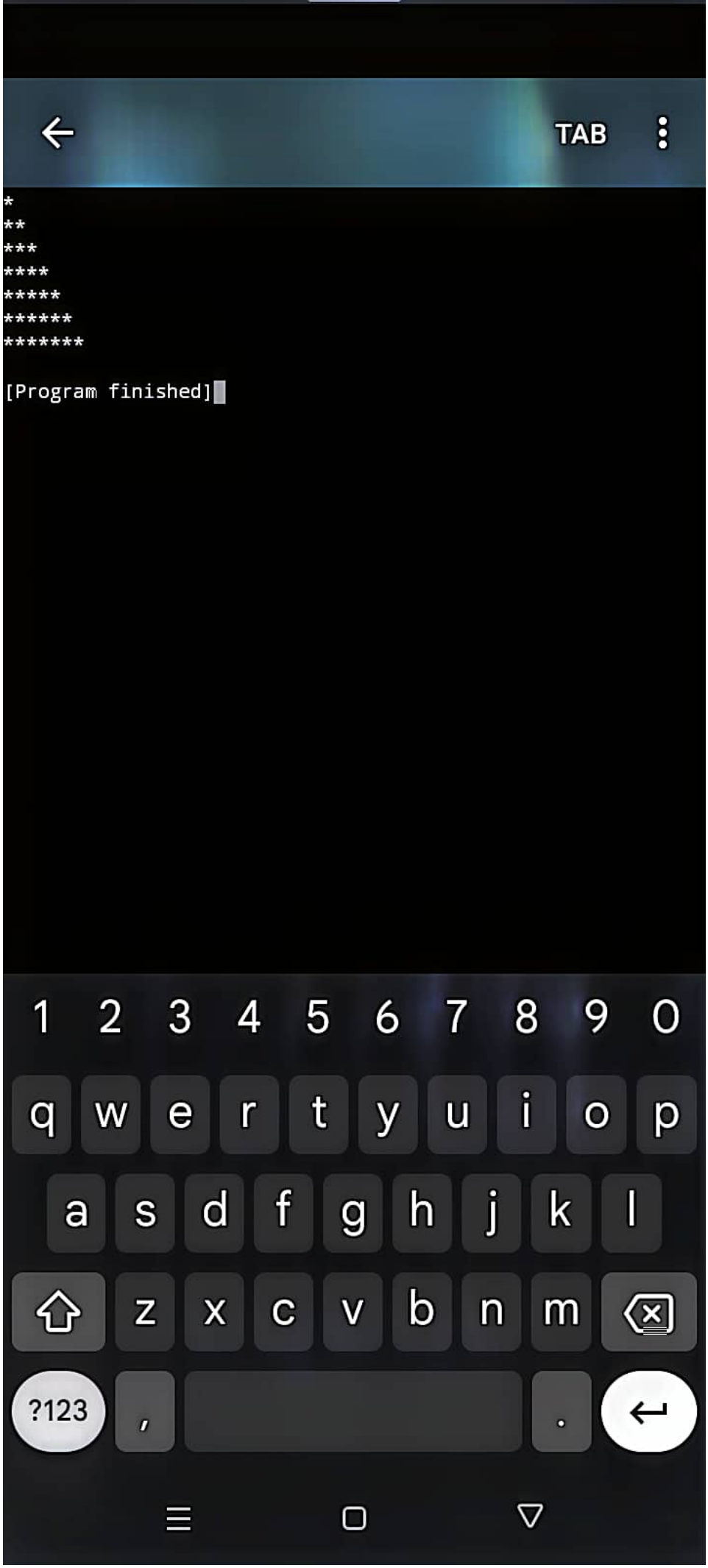
Input:-

```
# include <iostream>
  using namespace std;
  int main()
{
  for (int i =1 ; i<=7;i++)
 {
  for (j =1; j<=i; j++)
  {
    cout << "*";
  }
    cout << endl;
  }
    return 0;
 }
```

Output:-

```
*
**
***
****
*****
******
*******
```

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